Markscheme

Mathematics: analysis and approaches
Practice question bank

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32. (a)

Consider the function $f(x) = \dfrac{20x^2 - 24x - 14}{(2x-3)(x+1)(x-2)}$.

Write $f(x)$ as the sum of partial fractions in the form $\dfrac{A}{2x-3} + \dfrac{B}{x+1} + \dfrac{C}{x-2}$.

attempt to write $A(x+1)(x-2) + B(2x-3)(x-2) + C(2x-3)(x+1)$ *(M1)*

$$20x^2 - 24x - 14 = A(x+1)(x-2) + B(2x-3)(x-2) + C(2x-3)(x+1)$$ **A1**

substituting $x = \frac{3}{2}$ (the root of the non-monic factor $2x-3$) *(M1)*

$$-5 = -1.25A \Rightarrow A = 4$$ **A1**

substituting $x = -1$: $30 = 15B \Rightarrow B = 2$ **A1**

substituting $x = 2$: $18 = 3C \Rightarrow C = 6$ **A1**

[6 marks]

(b)

Hence, by differentiating your partial fraction form term by term, find $f'(0)$.

differentiate each term *(M1)*

$$f'(x) = -\frac{8}{(2x-3)^2} - \frac{2}{(x+1)^2} - \frac{6}{(x-2)^2}$$ **A1**

$$f'(0) = -\frac{8}{9} - 2 - \frac{3}{2} = -\frac{79}{18} \approx -4.389$$ **A1**

[3 marks]